

DUPRE AGCM, AB, Canada

WMO: 714140

Lat: 54.367N

Lon: 110.868W

Elev: 1824

StdP: 13.75

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-29.0	-23.6	-35.8	0.8	-28.5	-29.9	1.1	-23.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	82.0	63.2	79.0	62.1	76.1	60.8	66.9	76.9	64.7	74.8	62.9	72.4	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	92.1	72.6	60.8	85.3	69.5	59.0	79.9	67.1	32.5	77.0	30.7	75.0	29.3	72.6	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-38.3	87.1	4.5	2.2	-41.5	88.7	-44.2	89.9	-46.7	91.1	-49.9	92.7		
(5)			-38.5	71.4	4.5	2.6	-41.7	73.3	-44.3	74.8	-46.9	76.3	-50.1	78.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.4	8.1	7.4	21.3	36.7	51.2	58.6	62.1	59.9	50.9	38.0	21.0	7.8		
	DBStd	23.14	15.99	15.02	14.00	8.93	8.27	5.61	4.58	5.41	8.07	8.33	12.03	14.59		
	HDD50	6531	1300	1194	888	406	88	9	0	3	85	380	870	1308		
	HDD65	10853	1765	1614	1353	849	432	201	114	170	426	836	1320	1773		
	CDD50	1201	0	0	0	7	124	266	374	310	111	9	0	0		
	CDD65	48	0	0	0	0	3	8	23	12	2	0	0	0		
	CDH74	1091	0	0	0	4	135	169	353	292	135	3	0	0		
	CDH80	176	0	0	0	0	24	21	60	39	32	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	17.5	0.7	0.6	0.7	1.2	1.7	3.2	3.3	2.2	1.5	0.9	0.7	0.7		
(15) Precipitation	PrecMax	23.0	1.4	1.1	1.5	3.6	3.5	7.0	6.7	5.8	3.6	2.4	2.2	1.7		
	PrecMin	11.3	0.1	0.2	0.1	0.0	0.2	1.1	1.5	0.8	0.4	0.2	0.0	0.2		
	PrecStd	2.9	0.4	0.3	0.4	0.7	0.9	1.5	1.2	1.1	0.9	0.5	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.4	44.0	53.9	72.2	83.1	82.7	85.1	83.8	84.8	71.2	51.7	41.8		
		MCWB	36.9	37.2	42.4	50.5	57.4	61.4	66.9	64.9	62.3	53.5	40.2	35.0		
	2%	DB	40.5	38.3	47.3	63.3	77.9	78.5	81.4	80.3	78.1	61.3	44.5	36.0		
		MCWB	35.0	33.7	38.9	46.3	55.0	60.6	66.3	64.4	59.6	49.3	38.5	31.5		
	5%	DB	36.4	34.1	42.8	57.9	73.0	75.3	77.7	77.3	72.2	56.0	39.7	31.9		
		MCWB	32.9	30.5	36.2	43.5	53.3	59.3	64.0	63.5	57.3	45.9	35.8	28.6		
	10%	DB	32.1	30.8	38.9	52.2	68.8	72.1	74.8	74.3	66.6	51.7	36.2	27.9		
		MCWB	29.1	28.5	33.9	40.9	51.0	57.6	62.2	61.2	54.5	43.4	33.1	25.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.3	37.5	43.0	51.0	59.6	65.3	71.3	69.8	63.1	55.0	43.5	35.9		
		MCDB	43.7	42.1	51.7	71.2	75.5	76.6	80.9	79.9	78.3	67.5	49.9	41.0		
	2%	WB	35.4	33.8	39.0	46.7	57.0	63.1	67.9	66.7	60.7	50.3	39.3	31.8		
		MCDB	39.3	37.6	46.4	61.0	73.0	74.0	77.7	76.6	76.4	58.5	44.1	35.7		
	5%	WB	32.8	31.3	36.6	44.3	54.8	61.4	65.5	64.2	58.1	47.1	35.9	29.0		
		MCDB	36.5	34.0	42.0	57.0	68.9	71.2	75.3	74.1	70.5	54.7	39.0	31.7		
	10%	WB	29.1	28.2	34.3	41.7	52.7	59.6	63.6	62.3	55.6	44.2	33.4	25.6		
		MCDB	32.3	30.9	38.4	50.7	65.7	68.4	72.5	72.1	64.9	50.9	36.0	27.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.3	21.2	20.3	20.0	24.8	21.2	22.0	23.8	23.8	18.3	15.4	16.6		
		MCDBR	21.1	23.4	22.9	30.5	32.3	27.3	27.7	30.0	34.7	27.1	19.5	20.7		
		MCWBR	16.8	19.1	15.3	16.1	13.7	12.3	14.4	15.4	17.2	16.0	13.8	17.1		
		MCDBR	20.9	22.1	21.1	28.2	28.8	23.2	24.8	27.1	32.4	25.8	17.9	19.0		
	5% WB	MCWBR	17.1	18.5	14.3	15.5	13.0	12.2	13.7	14.9	16.8	15.6	13.4	15.9		
(40) Clear-Sky Solar Irradiance	taub		0.252	0.274	0.283	0.331	0.347	0.364	0.362	0.369	0.311	0.279	0.251	0.227		
	taud		2.284	2.261	2.375	2.377	2.388	2.350	2.408	2.341	2.513	2.574	2.454	2.318		
	Ebn at Noon		228	261	288	285	285	280	279	269	275	264	234	218		
	Edh at Noon		17	26	29	33	35	37	34	34	25	19	15	14		
(44) All-Sky Solar Radiation	RadAvg		276	605	1090	1477	1804	1848	1879	1531	1075	599	279	193		
	RadStd		29	49	78	123	150	128	102	121	115	96	36	19		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page